IKIMINATRACK: DIGITAL SAVINGS GROUP LEDGER AND MICRO-LOAN SYSTEM

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· GitHub: github.com/akandwanaho-joshua/joshua_20251SEN321_IkiminaTrack

THE PROBLEM WITH PAPER LEDGERS

· Community savings groups (Ikimina) are vital for financial inclusion in Rwanda.

· The Issue: Most rely on paper notebooks to track savings, loans, and interest.

· Who is affected: Members and treasurers.

· Why it matters: Paper records lead to human error, lost data, calculation mistakes, and financial disputes among members.

WHY I CHOSE THIS PROJECT

Financial inclusion is a major driver of economic growth in Rwanda.

Small savings groups are often ignored by expensive, large-scale banking software.

I want to create a scalable, automated database solution that brings professional financial tracking to the grassroots level.

PROJECT OBJECTIVES

Main Objective: To develop an automated Oracle database system using PL/SQL for managing community savings groups.

Specific Objectives:

1. Automate tracking of contributions and loan repayments.

2. Enforce business rules using PL/SQL triggers (e.g., blocking defaulters).

3. Automatically calculate interest and penalties.

4. Provide a real-time dashboard for the group treasurer.---

EXISTING SOLUTIONS & GAPS

Existing Systems: Musoni System, Odoo Accounting.

Gap 1: These systems are too expensive and overly complex for a local Ikimina.

Gap 2: They require internet and technical staff to maintain.

Our Solution: A lightweight, database-driven system where PL/SQL enforces the rules locally.

REAL DATA SOURCE

Source: Kaggle (Lending Club Microfinance Dataset).

Size: Over 2,000 records of anonymized loan data.

Licence: Open Database License (ODbL) - safe for educational use.

Adaptation: Interest rates and loan amounts will be scaled to reflect Rwandan Francs (RWF) and local microfinance standard

IKIMINATRACK & THE INNOVATION

· The Solution: An Oracle Database to centralize all member and financial data.

The Innovation: A Microsoft Power BI Dashboard connected directly to Oracle.

What it does: Allows the treasurer to view KPIs (total liquidity, default risks) and monthly trends.

Key Rule: PL/SQL enforces the rules; Power BI is just the visual interface.

PLANNED DATABASE DESIGN

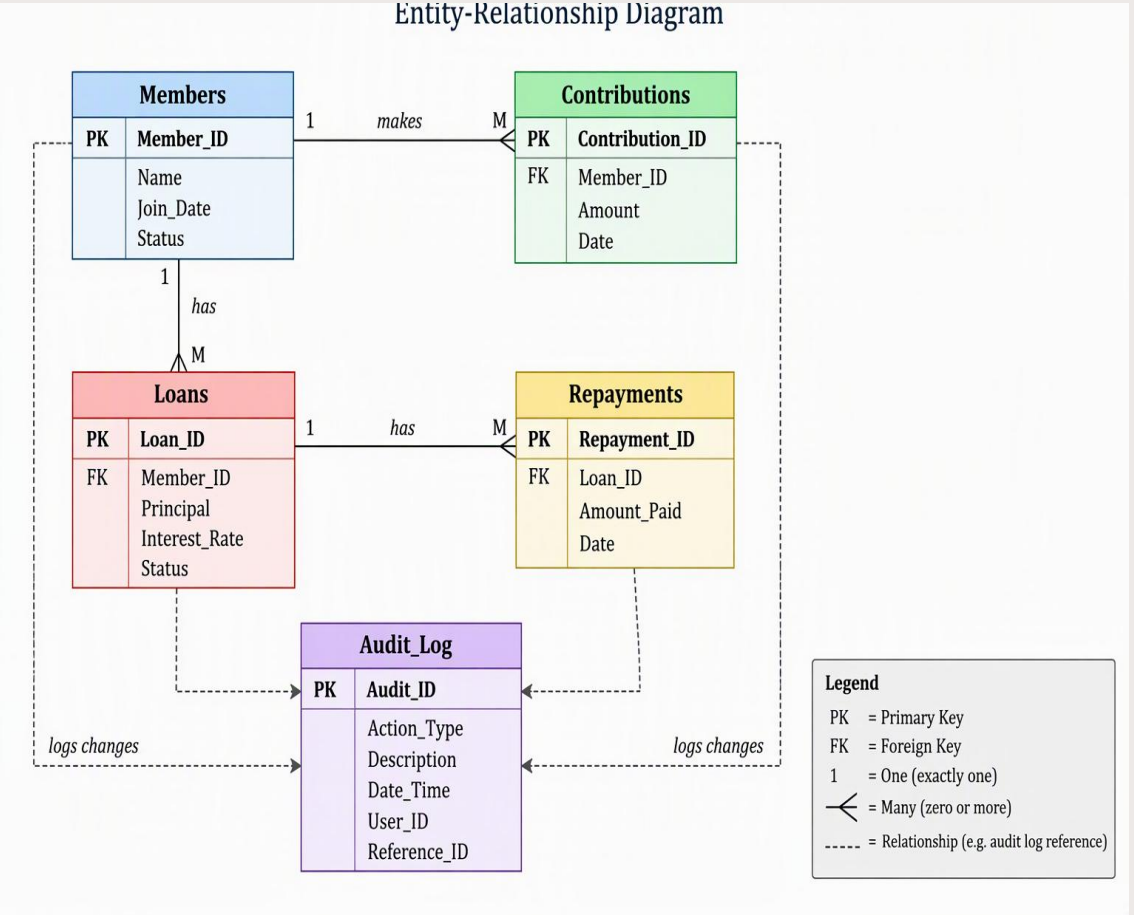
Members (Member_ID, Name, Join_Date, Status)

Contributions (Contribution_ID, Member_ID, Amount, Date)

Loans (Loan_ID, Member_ID, Principal, Interest_Rate, Status)

Repayments (Repayment_ID, Loan_ID, Amount_Paid, Date)

Audit_Log (Tracking changes and denied actions)



PL/SQL APPROACH

Trigger: Blocks a new loan if the member has a pending default or unpaid penalty.

Stored Procedure: record_contribution() auto-splits payment between savings and loan interest.

Function: calculate_outstanding_balance() returns a member's total debt.

Cursor: Loops through active loans at month-end to apply accrued interest.

Package: pkg_financials groups the loan and savings logic.---

EXPECTED RESULTS

A fully normalized Oracle PDB (Pluggable Database) with real data loaded.

Working PL/SQL scripts that successfully block invalid financial transactions.

A live demonstration of the Power BI dashboard showing real-time financial insights.

A secure audit log catching unauthorized actions.

FUTURE IMPROVEMENTS & LIMITATIONS

Limitation: The data is simulated from Kaggle, not direct live bank data.

Limitation: The Power BI interface is a prototype, not a full mobile app.

Future Work: Integrate Mobile Money (MoMo) APIs for automated SMS contribution logging.

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